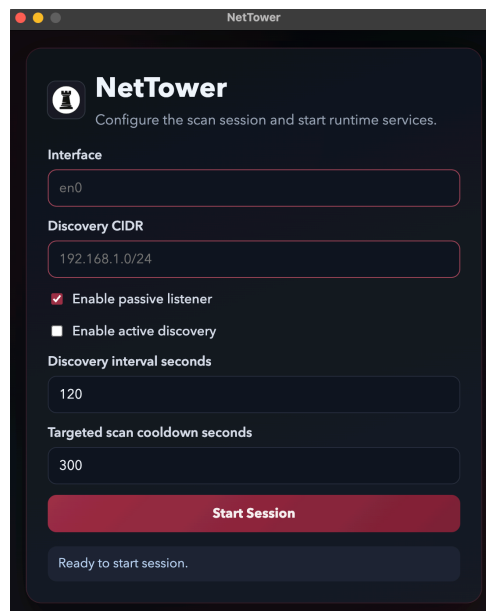


NetTower User Guide

Application: NetTower

Purpose: End-user guide for launching sessions, using topology views, controlling discovery settings, and troubleshooting.

Version Scope: Current Electron frontend and Python backend runtime behavior.



1) WHAT NETTOWER DOES

NetTower is an agentless network discovery and topology application.

It observes and infers devices and communication behavior, then renders live topology.

Core capabilities:

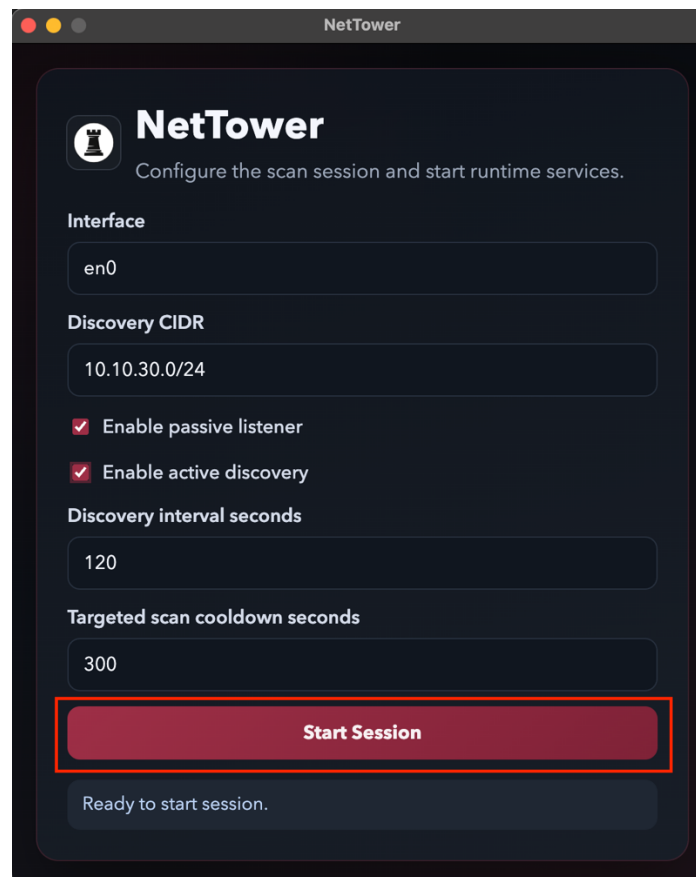
- Network topology visualization (2D + optional 3D + Idle wallboard mode)
- Node-centric exploration (click nodes for details)
- Host classification and role inference
- Connectivity relationship rendering (edges)
- Activity heat mapping by recency

- Dynamic updates as network state changes
- Interactive navigation (pan/zoom/camera)
- User-controlled scanning and discovery settings

2) SESSION LIFECYCLE

Startup flow:

1. Open NetTower.
2. Fill all required launch fields.
3. Press Start Session.
4. NetTower starts backend runtime and waits for readiness.
5. Dashboard opens when backend is ready.



The screenshot shows the NetTower application window. At the top, the title bar says "NetTower". Below it is a dark-themed interface. On the left is a circular icon with a white chess knight. To its right is the text "NetTower" in a large, bold, white font, followed by the subtitle "Configure the scan session and start runtime services." in a smaller white font. Below this, there are several configuration sections, each with a label in white and a corresponding input field or checkbox. The "Interface" section has a text input field containing "en0". The "Discovery CIDR" section has a text input field containing "10.10.30.0/24". Below these are two checkboxes, both checked with red checkmarks: "Enable passive listener" and "Enable active discovery". The "Discovery interval seconds" section has a text input field containing "120". The "Targeted scan cooldown seconds" section has a text input field containing "300". At the bottom of the configuration area is a large, rectangular, red button with the text "Start Session" in white. Below the button is a status bar with the text "Ready to start session." in a light gray font.

NetTower

 **NetTower**
Configure the scan session and start runtime services.

Interface

en0

Discovery CIDR

10.10.30.0/24

☒ Enable passive listener

☒ Enable active discovery

Discovery interval seconds

120

Targeted scan cooldown seconds

300

Start Session

Ready to start session.

Shutdown flow:

1. Press Stop Session in the dashboard top bar.
2. NetTower requests clean backend shutdown.
3. Application returns to launch screen.



3) LAUNCH CONFIGURATION SCREEN

All fields are required.

Field: Interface

- Description: Network interface used for observation and discovery.
- Validation: Required, interface format only.

Field: Discovery CIDR

- Description: Target range for interval discovery (example: 10.10.30.0/24).
- Validation: Required, valid IPv4 CIDR.

Field: Enable Passive Listener

- Description: Enables passive data collection from observed traffic.
- Behavior: ON by default.

Field: Enable Active Discovery

- Description: Enables probe-based discovery.
- Behavior: OFF by default unless user enables it.

Field: Discovery Interval Seconds

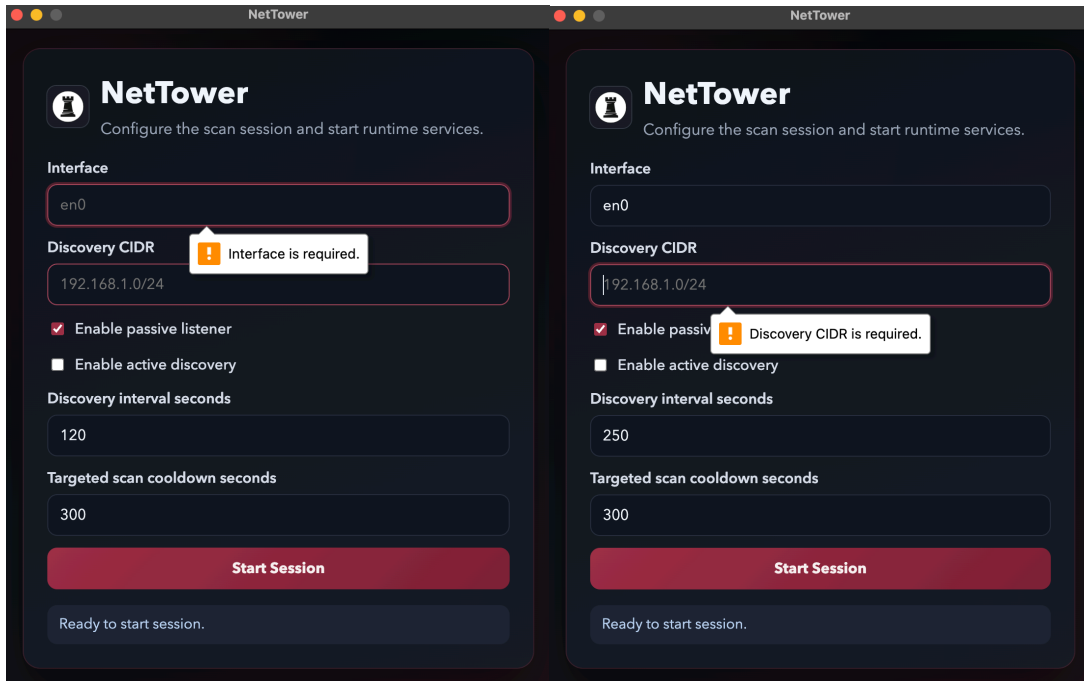
- Description: How often interval discovery runs.
- Validation: Required integer ≥ 1 .

Field: Targeted Scan Cooldown Seconds

- Description: Minimum delay between targeted scans of the same host.
- Validation: Required integer ≥ 0 .

Important validation behavior:

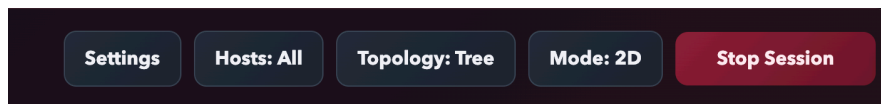
- At least one discovery mode must be enabled.
- Invalid interface/CIDR/numeric values block Start Session.



4) MAIN DASHBOARD OVERVIEW

Top bar controls:

- Settings
- Hosts: All / Public / Private
- Topology: Tree / Star
- Mode: 2D / 3D / Idle
- Stop Session



Activity heat legend:

- Active (0-30s): Green
- Warm (30-120s): Amber
- Cold (120s+): Red



5) TOPOLOGY VIEWS AND MODES

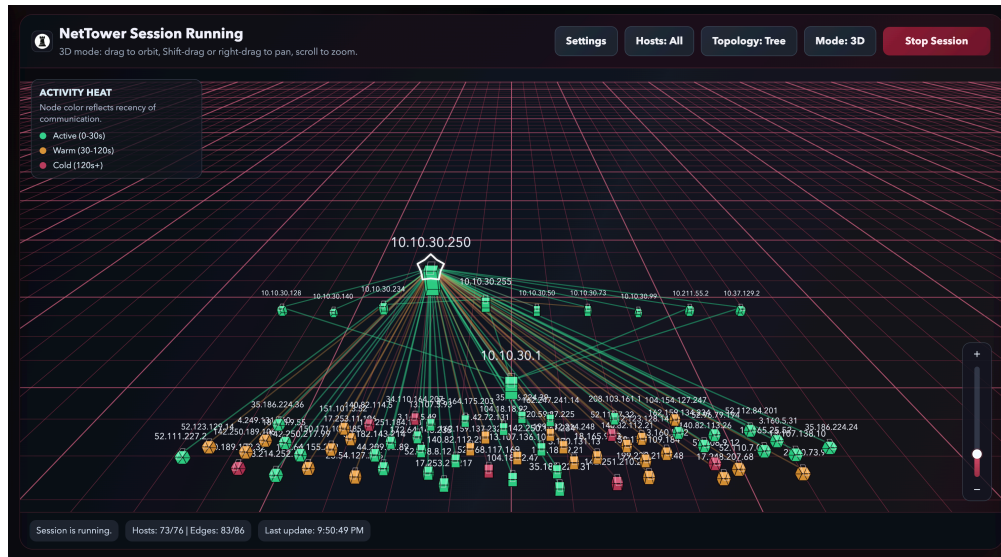
Mode: 2D

- Primary analysis mode
- Supports panning and zooming



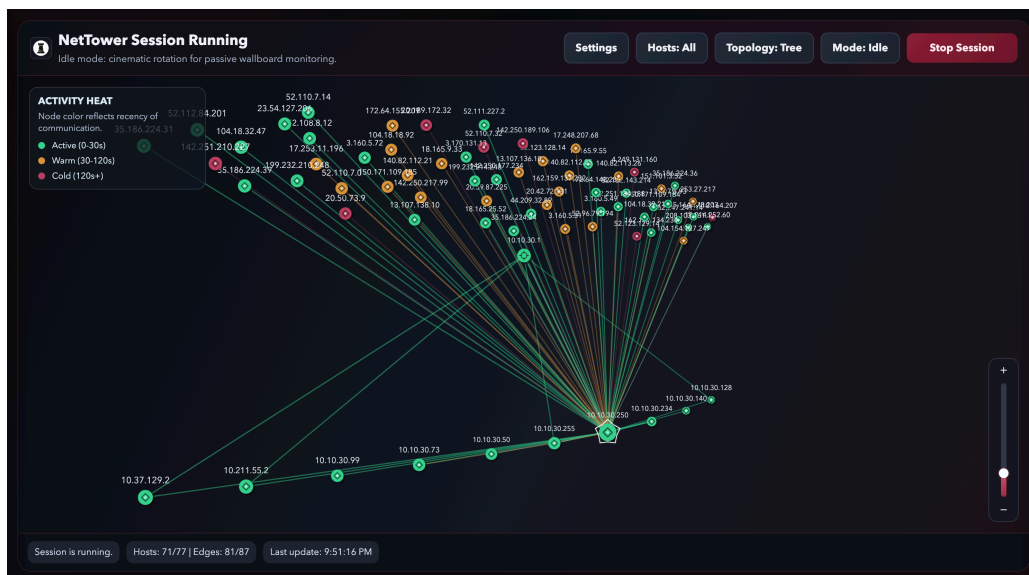
Mode: 3D

- Perspective view with interactive camera controls
- Supports orbit, pan, and zoom camera movement



Mode: Idle

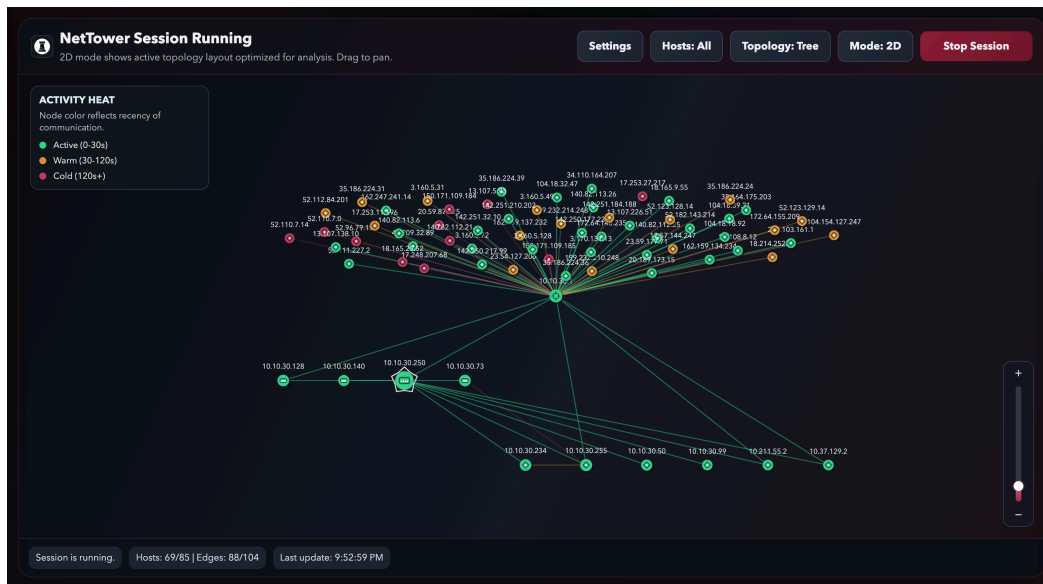
- Continuous rotating display suitable for monitor/wallboard view



6) TOPOLOGY LAYOUT OPTIONS

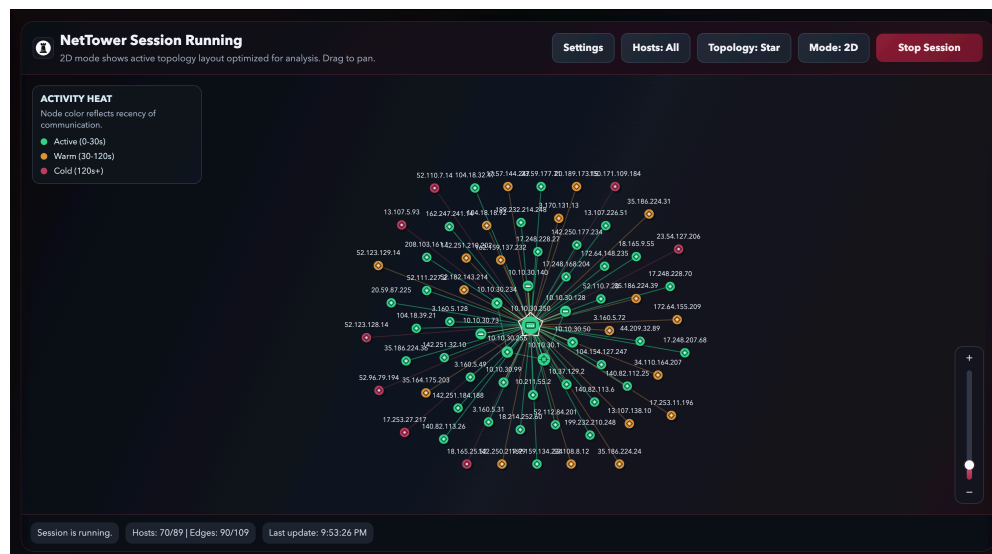
Layout: Tree (default)

- Top-down hierarchy of inferred network structure
- Intended to separate upstream/public, gateway/router, and internal hosts



Layout: Star

- Relationship-centric layout around key hosts
- Good for quickly seeing direct communication patterns



7) NODE-CENTRIC EXPLORATION

Each host is a selectable node.

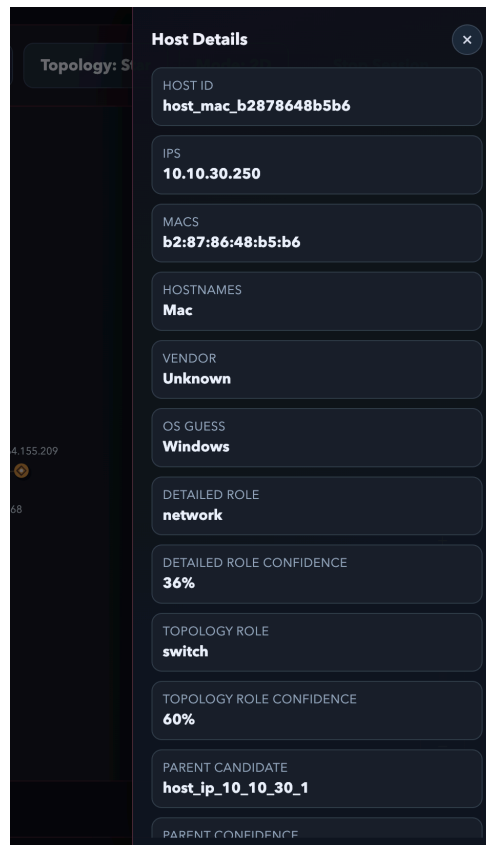
Clicking a node opens a right-side host details panel.

Host details include:

- Host ID
- IP addresses
- MAC addresses
- Hostnames
- Vendor
- OS guess
- Detailed role + confidence
- Topology role + confidence
- Parent candidate + confidence
- Topology layer
- First seen / Last seen
- Ports and services (when available)

Special marker:

- The machine running NetTower is visually marked for identification.



8) CONNECTIVITY REPRESENTATION

Observed or inferred relationships are rendered as edges.

Edge data updates as new traffic/discovery events are processed.

Tree mode uses inferred relationship metadata to improve hierarchy readability.



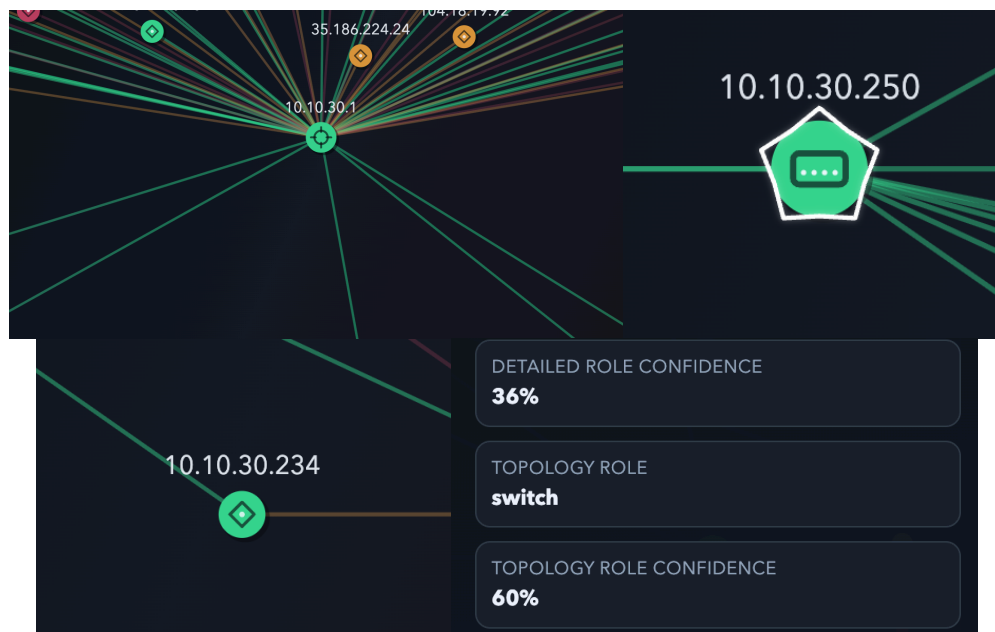
9) HOST IDENTIFICATION AND CLASSIFICATION

NetTower differentiates hosts using inferred role categories:

- Router/Gateway
- Switch/Network Device
- Server
- Workstation/Client
- Other/Unknown

Important:

- Role inference is probabilistic, not guaranteed ground truth.
- Confidence values are shown in host details to indicate certainty.



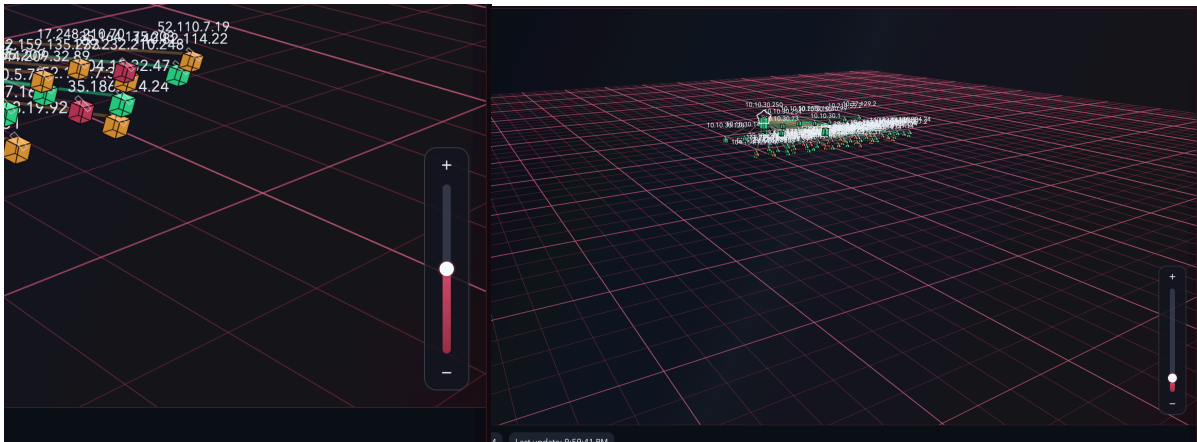
10) INTERACTIVE NAVIGATION CONTROLS

2D controls:

- Drag to pan view
- Vertical zoom slider at bottom-right
- Plus at top / Minus at bottom
- Red fill indicates zoom level on slider

3D controls:

- Drag: orbit camera
- Shift-drag or right-drag: pan camera
- Mouse wheel: zoom camera



11) SETTINGS PANEL

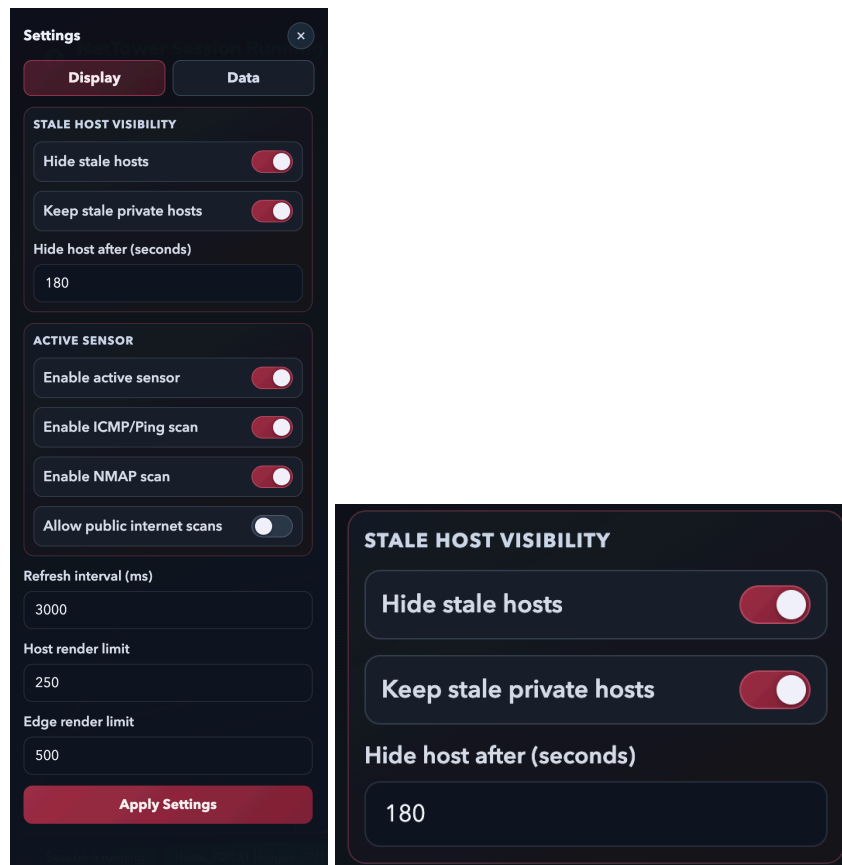
Settings includes at least:

- Display tab
- Data tab

11.1 Display Settings

- Hide stale hosts (default ON)
- Hide host after N seconds (editable when stale hiding is ON)

- Keep stale private hosts (dependent on stale host setting)
- Refresh interval
- Host render limit
- Edge render limit

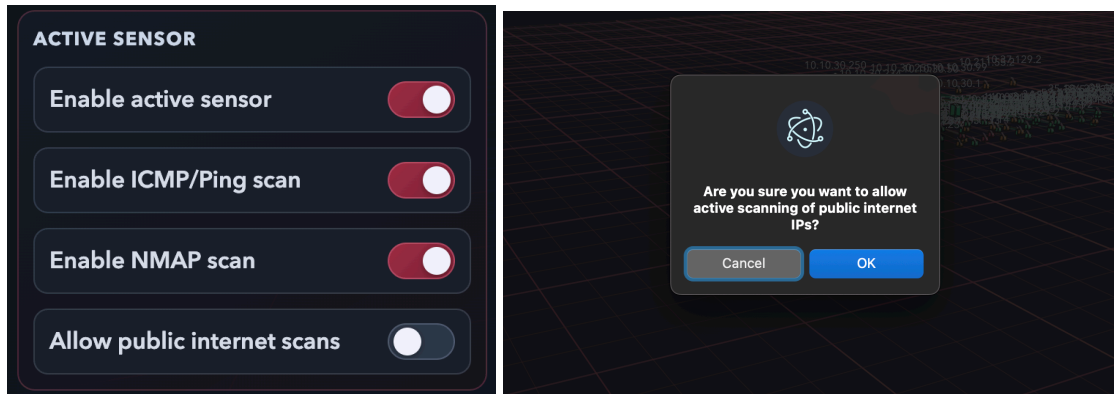


11.2 Active Discovery Settings

- Enable active sensor
- Enable ICMP/Ping scan (only interactable when active sensor ON)
- Enable NMAP scan (only interactable when active sensor ON)
- Allow public internet scans (confirmation prompt required)

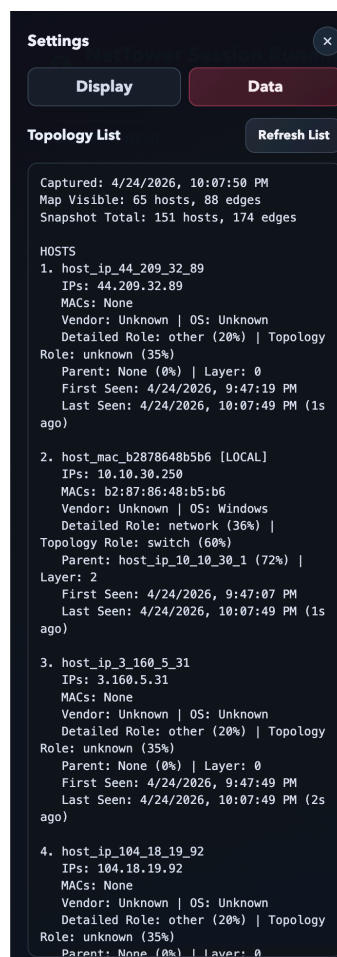
Safety behavior:

- Public active scanning is restricted by default.
- **Enable only with proper authorization.**



11.3 Data Tab

- Clean text listing of current hosts and edges
- Refresh control for latest data snapshot



12) SCANNING MODEL (USER LEVEL)

Passive discovery:

- Agentless observation from network traffic
- Useful for discovering active relationships naturally seen on the network

Active discovery:

- Probe-based scanning when enabled
- Can use ICMP, NMAP, or both (depending on toggles)
- Scope controlled by launch and settings configuration

Combined operation:

- Passive and active can run together
- Passive preserves traffic-observation context
- Active adds additional host/service signals when enabled

13) DYNAMIC UPDATES AND HEAT MAPPING

NetTower continuously refreshes topology state:

- New hosts appear as they are observed or confirmed
- Inactive hosts age and can be hidden based on stale settings
- Node color updates according to communication recency
- Edges update as relationships are observed/inferred

FIRST SEEN

4/24/2026, 10:07:30 PM

LAST SEEN

4/24/2026, 10:11:25 PM

Session is running.

Hosts: 66/169 | Edges: 88/193

Last update: 10:13:56 PM

14) TROUBLESHOOTING

Issue: Failed to start session / Python bridge exited

- Verify Python environment and backend dependencies.
- Check backend/runtime logs for startup exceptions.

Issue: Stuck waiting for backend readiness

- Backend may not have created readiness flag yet.
- Check runtime and backend logs.

Issue: "nmap not found"

- Nmap is not installed or not in the PATH used by backend runtime.
- Install Nmap and confirm the executable is discoverable before launching app.

Issue: No edges or very few hosts shown

- Verify passive listener is enabled.
- Check render limits and host filters.
- Confirm active settings are not overly restrictive.

Issue: Hosts unexpectedly disappear

- Check stale host hiding threshold and stale-private toggle behavior.

```
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.runtime.supervisor | Supervisor initialized (idle state)
2026-04-24 22:14:56 | INFO | backEnd.runtime.supervisor | Starting NetTower session
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.runtime.mongo_runtime | Removing previous Mongo data dir: /Users/benmolloy/Documents/NetTower-Capstone-Project/code/runtime/mongo_data
2026-04-24 22:14:56 | INFO | backEnd.runtime.mongo_runtime | Starting local mongod: host=127.0.0.1 port=27017 dbpath=/Users/benmolloy/Documents/NetTower-Capstone-Project/code/runtime/mongo_data binary=/Users/benmolloy/Documents/NetTower-Capstone-Project/code/runtime_binaries/macos/mongod
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.runtime.mongo_runtime | mongod is ready on 127.0.0.1:27017
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.runtime.supervisor | Launching backend: ['/Users/benmolloy/Documents/NetTower-Capstone-Project/code/venv/bin/python', '-m', 'backEnd.main']
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | NetTower backend starting...
2026-04-24 22:14:56 | INFO | backEnd.main | Interface: en0
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Default gateway: 10.10.30.1
2026-04-24 22:14:56 | INFO | backEnd.main | Discovery CIDR: 10.10.30.1/24
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Active discovery scope: private-only (RFC1918)
2026-04-24 22:14:56 | INFO | backEnd.main | Active discovery methods: icmp=True nmap=True
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Mongo client connected.
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Cleared Mongo collections for new session: hosts=172, edges=196
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Seeded local host into topology: 10.10.30.250
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Seeded default gateway into topology: 10.10.30.1
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.main | Passive listener started
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.sensors.passive_listener | Starting passive capture backend=tcpdump iface=en0 filters='arp or ip' cmd=['/usr/sbin/tcpdump', '-l', '-n', '-e', '-i', 'en0', 'arp or ip']
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.sensors.passive_listener | passive listener started on iface=en0 backend=tcpdump filters='arp or ip'
[NetTower bridge] 2026-04-24 22:14:56 | WARNING | backEnd.sensors.passive_listener | tcpdump stderr: tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
2026-04-24 22:14:56 | WARNING | backEnd.sensors.passive_listener | tcpdump stderr: listening on en0, link-type EN10MB (Ethernet), snapshot length 524288 bytes
[NetTower bridge] 2026-04-24 22:14:56 | INFO | backEnd.runtime.supervisor | Backend ready: /Users/benmolloy/Documents/NetTower-Capstone-Project/code/runtime/backend_ready.flag
```

15) QUICK WORKFLOWS

Workflow A: Monitor local network safely

1. Enable passive listener.
2. Keep active discovery OFF initially.
3. Start session and inspect tree view.
4. Use host filter and details drawer to validate known devices.

Workflow B: Add active context safely

1. Enable active sensor in settings.
2. Keep public scanning OFF.
3. Use ICMP and NMAP toggles based on need.
4. Observe new enrichment in host details.

Workflow C: Export-readable view for review

1. Open Settings -> Data tab.
2. Refresh host/edge list.
3. Capture text output and screenshots for reports.

16) TERMS

Host:

- Discovered network entity represented as a node.

Edge:

- Observed or inferred relationship between hosts.

Passive Discovery:

- Traffic observation without direct probes.

Active Discovery:

- Direct probing (ICMP and/or NMAP depending on settings).

Stale Host:

- Host not observed within configured age threshold.

Role Confidence:

- Confidence value for inferred role labels.